



Aryabrata Basu

Curriculum Vitae

"Science is built up of facts, as a house is built of stones; but an accumulation of facts is no more a science than a heap of stones is a house." – Henri Poincaré, Science and Hypothesis, 1905

Current Positions

- 2022–present **Assistant Professor (Tenure Track)**, *Department of Computer Science, University of Arkansas at Little Rock, Little Rock, AR, United States*
- 2022–present **Research Fellow**, *Donaghey Emerging Analytics Center, University of Arkansas at Little Rock, Little Rock, AR, United States*
- 2025–present **Undergraduate Program Coordinator**, *Department of Computer Science, University of Arkansas at Little Rock, Little Rock, AR, United States*

Past Positions

- 2016–2022 **Visual Information Specialist**, *Emory Center for Digital Scholarship, Emory University, Atlanta, GA, United States*
- 2010–2016 **Graduate Research Assistant**, *University of Georgia, Athens, GA, United States*
- 2009–2010 **Graduate Teaching Assistant**, *University of Georgia, Athens, GA, United States*

Education

- 2009–2018 **Ph.D. - Computer Science**, *University of Georgia, Athens, GA, United States*
- 2003–2007 **Bachelor of Technology - Information Technology**, *West Bengal University of Technology, Kolkata, West Bengal, India*

PhD Dissertation

- Dissertation [Designing and evaluating ubiquitous wearable virtual reality](#)
- Advisor(s) Johnsen K, Potter D, Rasheed K, and Liu T

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Research Interests

Topics Human-Computer Interaction, Virtual Reality + Augmented Reality = Extended Reality, Virtual Worlds, Virtual Avatars, Animation, Natural User Interfaces (2D/3D), Usability Analytics, Study Design, Computer Graphics, Artificial Intelligence, Machine Learning, Distributed Computing, Robotics, Prototyping wearable hardware and its design, and Haptics.

Honors and Recognition

- 2025 **Best Paper Award ('Honorable Mention')** - 'From Novelty to Knowledge: A Longitudinal Investigation of the Novelty Effect on Learning Outcomes in Virtual Reality', IEEE Virtual Reality.
- 2014 **Best Paper Award** - 'Mixed Reality Virtual Pets to Reduce Childhood Obesity', IEEE Virtual Reality.
- 2013 **Best Poster Award** - 'Evolution and Usability of Ubiquitous Immersive 3D Interfaces', IEEE Symposium on 3D User Interfaces.
- 2010 **Best Poster Award ('Honorable Mention')** - 'Field Valid Probability Assessment in Uncertain and Multi-agent Settings', Research Day 2010, Department of Computer Science, University of Georgia.

Media and Public Engagement

- 2025 **America250 – [Our American Story](#)**, Featured Interviewee (sole researcher profile from the State of Arkansas). Library of Congress archival project; filmed at Old State House, Little Rock, AR. Oct 2025.
- 2025 **KUAR 89.1 (NPR member station), Morning Edition** — Guest interview on [Mapping cardiac blood vessels in 3D](#). Air date: Sep 11, 2025.

Grants

- 2024-25 **Cybersecurity Consortium for Innovation** - US Department of Energy, National Energy Technology Laboratory, \$5.0 million in total funding, Role: Co-Investigator
- 2023-24 **Robust 2D to 3D Cardiac Arterial Mapping and Visualization for Surgical Preplanning** - National Science Foundation/EPSCoR-DART, \$100,000 in total funding, Role: Principal Investigator

Teaching

- 2025 **Professor**, *Capstone Projects (CPSI 49203), Adv Game Programming (CPSI 37703), Seminar in Computer Architecture (CPSI 73103)*, UA Little Rock, Arkansas
- 2024 **Professor**, *Computer Networks (CPSC 3384), Database Concepts (CPSC 3375), Computer Architecture (CPSC 7331), Distributed Computing (CPSC 43/5387)*, UA Little Rock, Arkansas

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2023 **Professor**, *Computer Networks (CPSC 3384), Database Concepts (CPSC 3375), Academic Computer Architecture (CPSC 7331), Distributed Computing (CPSC 43/5387)*, UA Little Rock, Arkansas

Fall 2022 **Professor**, *Distributed Computing (CPSC 4387/5387)*, UA Little Rock, Arkansas

2019 – 2022 **Lead Instructor**, *Introduction to 3D Visualization and Interactive Media Design (ARTHIST 393R)*, Emory University, Atlanta

Service

2025-26 **Program Committee Member**: IEEE International Conference on Artificial Intelligence & eXtended and Virtual Reality (AIxVR 2026)

2025 **Editorial Board Member**: [Nature Portfolio](#)

2025 **Track Chair(Main+Posters)**: The ACM Symposium on Applied Perception (SAP)

2025 **Program Committee Member**: TRUST-XR 2025 @IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

2024-25 **Program Committee Member**: IEEE International Conference on Artificial Intelligence & eXtended and Virtual Reality (AIxVR 2025)

2024 **Panel Reviewer**: [NEH](#)

2021 **University Senate Committee Member**: [COE](#)

2020 **Panel Reviewer**: [NSF](#)

2020 **Associate Editor**: [Presence](#)

2020 **Review Editor**: [Frontiers in Virtual Reality](#)

2020 **Review Editor**: [Springer Nature Virtual Reality](#)

2020-21 **Program Committee Member**: International Symposium on Visual Computing

2020 **Session Chair**: Immersive Learning Research Network

2020 **Local Arrangements Chair, Usability Study Researcher**: IEEE Virtual Reality

2012-2013 **Student Volunteer**: IEEE Virtual Reality

2012 **Student Volunteer**: IEEE International Symposium on Mixed and Augmented Reality

Membership

- IEEE Senior Member
- ACM Lifetime Member

Publications

Sunny, M. J. M., Springer, J. P., and Basu, A. Evaluating adjustment and proficiency disparities in virtual reality. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1282–1283, 2025.

Sunny, M. J. M., Sai Nikitha Guthula, J., Kharel, A., Shraboni, M. N., Rashid, H., Bhandari, P., Springer, J. P., and Basu, A. Gaze insights in xr: Real-time eye-tracking analytics with elasticsearch. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 97–103, 2025.

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- Nikitha Guthula, J. S., Rashid, H., Springer, J. P., and Basu, A. Preserving privacy in vr telemetry data. In *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1270–1271, 2025.
- Lee, J., Chen, C. C., and Basu, A. From novelty to knowledge: A longitudinal investigation of the novelty effect on learning outcomes in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 31(5):3204–3212, 2025.
- Tullis, J. A., Lanter, D. P., Basu, A., Cothren, J. D., Shi, X., Limp, W. F., Linck, R. F., Young, S. G., Davis, J., and Alsumaiti, T. S. Geoprocessing, workflows, and provenance. In *Remote Sensing Handbook, Volume II*, pages 303–337. CRC Press, 2024.
- Sunny, M. J. M. and Basu, A. Non-linear parameterization of spatial decision making in immersive virtual environment. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 389–395. IEEE, 2024.
- Guthula, J. S. N. and Basu, A. Navigating gender biases in xr: Towards equitable technological future. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 326–329. IEEE, 2024.
- Page, M., Kang, Y., Varner, S., Lee, K., Cors, A., Basu, A., Durante, K., and Burr, I. Methods and evaluation in the historical mapping of cities. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 48:155–160, 2023.
- Basu, A., Sunny, M. J. M., and Guthula, J. S. N. Privacy concerns from variances in spatial navigability in vr. *arXiv preprint arXiv:2302.02525*, 2023.
- Basu, A. Stag: A tool for realtime replay and analysis of spatial trajectory and gaze information captured in immersive environments. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 43–45. IEEE, 2022.
- Martin, A. J., Stearns, D., Whitten, M. J., Hage, M. M., Page, M., and Basu, A. First known trace fossil of a nesting iguana (pleistocene), the bahamas. *Plos One*, 2020.
- Basu, A. Work-in-progress—augmented reality at scale: Using physical dimensionality of spaces to our advantage! In *6th International Conference of the Immersive Learning Research Network (iLRN)*, pages 373–376. IEEE, 2020.
- Basu, A. Work-in-progress—tracking the untracked. *arXiv preprint arXiv:1909.05327*, 2019.
- Basu, A. Work-in-progress—a brief chronology of virtual reality. *arXiv preprint arXiv:1911.09605*, 2019.
- Basu, A. and Johnsen, K. Work-in-progress—navigating a maze differently-a user study. *arXiv preprint arXiv:1805.09454*, 2018.
- Basu, A., Ball, C., Manning, B., and Johnsen, K. Effects of user physical fitness on performance in virtual reality. In *IEEE Symposium on 3D User Interfaces (3DUI)*, pages 233–234. IEEE, 2016.
- Ahn, S. J., Johnsen, K., Robertson, T., Moore, J., Brown, S., Marable, A., and Basu, A. Using virtual pets to promote physical activity in children: An application of the youth physical activity promotion model. *Journal of Health Communication*, 20(7):807–815, 2015.
- Johnsen, K., Ahn, S. J., Moore, J., Brown, S., Robertson, T. P., Marable, A., and Basu, A. Mixed reality virtual pets to reduce childhood obesity. *Visualization and Computer Graphics, IEEE Transactions on*, 20(4):523–530, 2014.
- Basu, A. and Johnsen, K. Ubiquitous virtual reality ‘to-go’. In *IEEE Virtual Reality (VR)*, pages 161–162. IEEE, 2014.

- Basu, A., Johnsen, K., Bogert, K., and Wins, P. Immersive virtual reality on-the-go. In *Virtual Reality (VR)*, IEEE, pages 193–194. IEEE, 2013.
- Basu, A., Johnsen, K., and Bogert, K. Poster: Evolution and usability of ubiquitous immersive 3d interfaces. In *IEEE Symposium on 3D User Interfaces (3DUI)*, pages 131–132. IEEE, 2013.
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- Wins, P., Basu, A., and Johnsen, K. Do-it-yourself interface device prototyping for virtual reality. *International Journal of Virtual Reality*, 11(3):43–48, 2012.
- Basu, A., Saupe, C., Refour, E., Rajj, A., and Johnsen, K. Immersive 3dUI on one dollar a day. In *IEEE symposium on 3D user interfaces (3DUI)*, pages 97–100. IEEE, 2012.
- Basu, A., Rajj, A., and Johnsen, K. Ubiquitous collaborative activity virtual environments. In *Proceedings of the ACM 2012 conference on Computer Supported Cooperative Work*, pages 647–650. ACM, 2012.
- Mrázek, J., Chaudhari, T., and Basu, A. Perplot & perscan: tools for analysis of dna curvature-related periodicity in genomic nucleotide sequences. *Microbial informatics and experimentation*, 1(1):13, 2011.
- Basu, A. and Nachtegaal, M. Owa filters: A robust filtering method and its application to color images. In *IEEE Symposium on Computational Intelligence for Image Processing*, pages 8–13, 2009.
- Basu, A. and Funk, S. An optimal scheme for multiprocessor task scheduling: a machine learning approach. *Work-In-Progress Proceedings, The 30th IEEE Real-Time Systems Symposium*, page 29, 2009.